

6. Protease and lipase are enzymes that are added to biological washing powders to help break down stains on clothing.

- (a) Complete the sentences below by stating the **two** types of molecules that the enzymes in the biological washing powders help to break down. [2]

Protease breaks down

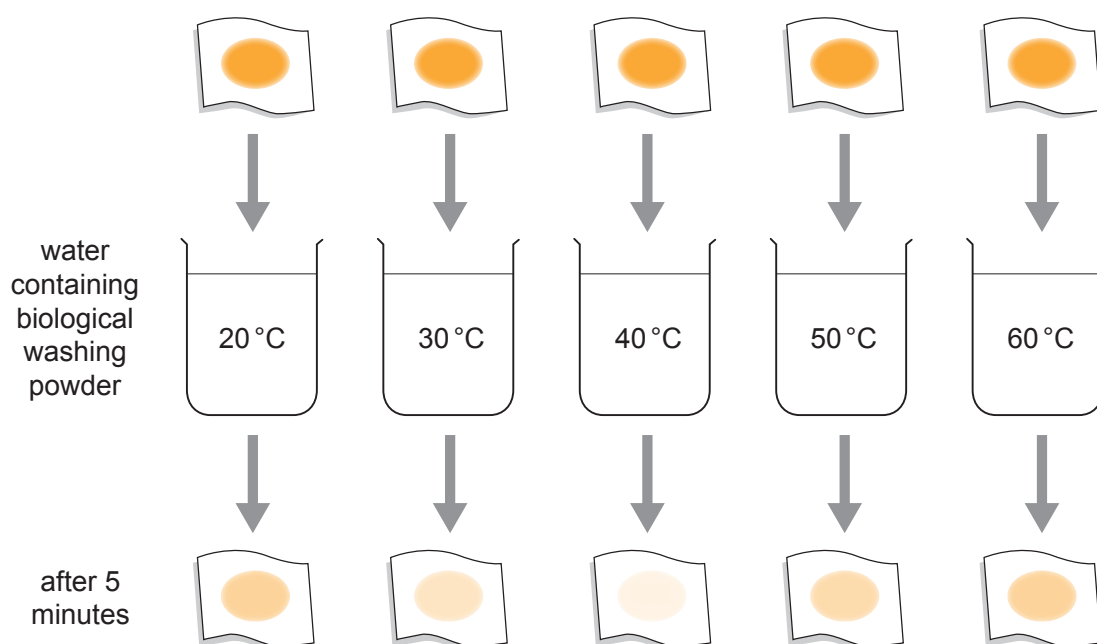
Lipase breaks down

- (b) Students designed an experiment to test the effect of temperature on the activity of these enzymes in biological washing powder. They used the following method:

- Stain five pieces of fabric with egg.
- Set up five beakers of water at 20 °C, 30 °C, 40 °C, 50 °C and 60 °C.
- Add biological washing powder to each beaker.
- Add a piece of stained fabric to each beaker and leave for 5 minutes.
- Remove the samples of fabric and compare the stain remaining.

The method and results of the experiment are shown in **Image 6.1**.

Image 6.1



- (i) Suggest the best temperature to wash clothes using this biological washing powder. Use the results in **Image 6.1** to give the reason for your answer. [2]

Temperature °C

Reason

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(ii) Use your knowledge of enzymes to explain the result at 60 °C. [3]

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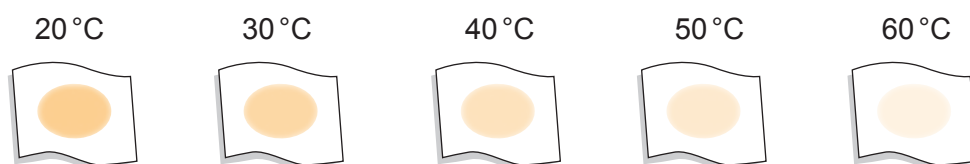
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(c) The students repeated the experiment using the same method but using non-biological washing powder which does not contain enzymes. Their results are shown in **Image 6.2**.

Image 6.2



(i) Describe the results with the non-biological washing powder. [1]

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(ii) State why they repeated the experiment using non-biological washing powder. [1]

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